

Chunggi Lee

PHD STUDENT @ HARVARD UNIVERSITY

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Education

Harvard University

PH.D IN COMPUTER SCIENCE

- GPA: 3.92/4.0, Advisor : [Prof. Hanspeter Pfister](#)

Boston, MA

Sep. 2023 - Now

Ulsan National Institute of Science and Technology

M.S IN ELECTRICAL AND COMPUTER ENGINEERING

- GPA : 4.17/4.3, Advisor : [Prof. Sungahn Ko](#)
- Thesis Title: GUIComp: A GUI Design Assistant with RealTime, MultiFaceted Feedback

Ulsan, South Korea

Mar. 2018 - Feb. 2020

Ulsan National Institute of Science and Technology

B.S IN ELECTRICAL AND COMPUTER ENGINEERING

- Magna Cum Laude
- Thesis Title: A Visual Analytics System for Exploring, Monitoring, and Forecasting Road Traffic Congestion.

Ulsan, South Korea

Mar. 2014 - Feb. 2018

Publications

INTERNATIONAL

- P1. **Chunggi Lee**, Tica Lin, Hanspeter Pfister, Zhu-Tian Chen, “**Sportify: Question Answering with Embedded Visualizations and Personified Narratives for Sports Video.**”, IEEE Transactions on Visualization and Computer Graphics ([IEEE VIS](#)), 2024. [[PDF](#)] [[Webpage](#)]
- P2. Namhyuk Ahn, Junsoo Lee, **Chunggi Lee**, Kunhee Kim, Daesik Kim, Seung-Hun Nam, Kibeom Hong, “**DreamStyler: Paint by Style Inversion with Text-to-Image Diffusion Models.**”, The Association for the Advancement of Artificial Intelligence ([AAAI](#)), 2024. [[PDF](#)] [[Webpage](#)] [[Code](#)]
- P3. Han Kim*, **Chunggi Lee***, Junsoo Lee*, Dohyun Kim, Kwangjin Lee, Moohyun Oh, Daesik Kim, “**FlatGAN: A Holistic Approach for Robust Flat-Coloring in High-Definition with Understanding Line Discontinuity.**”, ACM Multimedia ([MM](#)), 2023.
- P4. Cholmin Kang, **Chunggi Lee**, Heon Song, Minuk Ma, Sergio Pereira, “**Variability Matters : Evaluating inter-rater variability in histopathology for robust cell detection.**”, European Conference on Computer Vision Workshop On AI-Enabled Medical Image Analysis ([ECCVW](#)), 2022. [[PDF](#)]
- P5. **Chunggi Lee**, Seonwook Park, Heon Song, Jeongun Ryu, Sanghoon Kim, Haejoon Kim, Sergio Pereira, Donggeun Yoo, “**Interactive Multi-Class Tiny-Object Detection.**”, IEEE Conference on Computer Vision and Pattern Recognition ([CVPR](#)), 2022. [[PDF](#)] [[VIDEO](#)]
- P6. Cheonbok Park, **Chunggi Lee**, Hyojin Bahng, Yunwon Tae, Kihwan Kim, Seungmin Jin, Sungahn Ko, Jaegul Choo, “**ST-GRAT: A novel spatio-temporal graph attention networks for accurately forecasting dynamically changing road speed.**”, ACM International Conference on Information and Knowledge Management ([CIKM](#)), 2020 (20.9% acceptance rate). [[PDF](#)]
- P7. **Chunggi Lee**, Sanghoon Kim, Dongyun Han, Hongjun Yang, Young-Woo Park, Bum Chul Kwon, Sungahn Ko, “**GUIComp: A GUI Design Assistant with RealTime, MultiFaceted Feedback**”, ACM CHI Conference on Human Factors in Computing Systems ([CHI](#)), 2020, Accepted. [[PDF](#)] [[Preview Video](#)] [[VIDEO](#)]
- P8. **Chunggi Lee**, Yeonjun Kim, Seungmin Jin, Dongmin Kim, Ross Maciejewski, David Ebert, and Sungahn Ko, “**A Visual Analytics System for Exploring, Monitoring, and Forecasting Road Traffic Congestion.**” IEEE transactions on visualization and computer graphics ([TVCG](#) IF=4.579), 2019 (Proc. [IEEE VIS'19](#)), Accepted. [[LINK](#)] [[PDF](#)] [[VIDEO](#)][[NEWS1](#)] [[NEWS2](#)] [[NEWS3](#)]

PREPRINT, POSTER, DOMESTIC

- D1. Hyunwook Lee, **Chunggi Lee**, Hongkyu Lim, Sungahn Ko, “**TILDE-Q: A Transformation Invariant Loss Function for Time-Series Forecasting.**”, (Under Review), 2022.
- D2. Juyoung Oh, **Chunggi Lee**, Hwiyeon Kim, Kihwan Kim, Osang Kwon, Eric D. Ragan, Bum Chul Kwon, Sungahn Ko, “**An Empirical Study on the Relationship Between the Number of Coordinated Views and Visual Analysis.**”, Arxiv. [[PDF](#)]
- D3. **Chunggi Lee**, Juyoung Oh, Seungmin Jin, Isaac Cho, and Sungahn Ko, “**A Graphical Workflow Exploration Environment For Visual Analytics.**”, Arxiv. [[PDF](#)]
- D4. Kihwan Kim, Sanghoon Kim, **Chunggi Lee**, Sungahn Ko, “**Modeling Exploration/Exploitation Decisions through Mobile Sensing for Understanding Mechanisms of Addiction.**”, ([MobiSys](#)), 2019 (Poster)
- D5. J. Lee, K. Kim, **C. Lee**, and S. Ko, “Visualization based Deep Learning Analysis Technology.” The Korean Society for Noise and Vibration Engineering ([KSNVE](#)), 2017, Accepted. [[LINK](#)] [[PDF](#)]
- D6. Y. Oh, **C. Lee**, J. Oh, J. Yang, H. Gwag, S. Moon, S. Park, and S. Ko, “Introdcuton of Visual Analytics.” [Korea Computer Graphics Society](#), 2016, Accepted. [[LINK](#)] [[PDF](#)]

Professional Experience

Naver Webtoon AI.

Gyeonggi-do, South Korea

RESEARCHER SCIENTIST (ALTERNATIVE MILITARY SERVICE) [LINK]

Aug 2022 - Aug 2023

- Conducted research on **stable diffusion models** to support the process of webtoon (comics) creation
- Designed and built an **AI platform** frontend and backend to distribute deep learning models for users.

Lunit Inc.

Seoul, South Korea

RESEARCHER ENGINEER (ALTERNATIVE MILITARY SERVICE) [LINK]

Mar 2020 - Aug-2022

- Designed and built interactive deep learning models for reducing cost of annotations (e.g., **object detection and segmentation**). (P2)
- Implemented an annotation tool and an internal operation tool with **Django, DRF, React, and mui**.
- Deployed and served deep learning models by using **TorchServe and Fast API**.
- Analyzed quality of pathology data which have significant different among experts and proposed a new method for evaluating pathology large-scale dataset (P1)

Interactive Visual Analysis & Data Exploration Research Lab

Ulsan, South Korea

RESEARCHER & DEVELOPER [LINK]

Jun 2016 - Mar 2020

- Built interactive tools and visual analytic systems for **mobile GUI design authoring for novice** (P4), **traffic congestion forecasting and propagation** (P5), **multiple coordinate visualization** (D3), and **gene expression** (D4).
- Implemented **spatio-temporal deep learning models** (P3), mobile GUI recommendation deep learning, biomedical data clustering

Honors & Awards

2017	4th Place & Bronze Prize , National Super Computing Competition	Ulsan, South Korea & KISTI & UNIST
2017	3rd Place & Bronze Prize , Naver UNIST Undergraduate Poster Award '17 UNIST \$1,500 [POSTER]	Naver & UNIST
2016 & 2017	Semester GPA 3.94/4.3, GPA 3.93/4.3 , Semester Academic Excellence Award for 2016 Fall Semester, 2017 Fall Semester	UNIST
2016	3rd Place , HeXAThon, Develop Customization Interior Design VR Application, Received an award of \$500	Ulsan, South Korea & NAVER
2016	3rd Place , Competition of Using Public Data of Ministry of Trade, Industry and Energy, Develop Emotional Color Combination VR Application, Received an award of \$1,000	Seoul, South Korea & Ministry of Trade, Industry and Energy

Patent

- **Graphic User Interface Assistant Device and Method in Mobile Environment** – Application Number : 10-2019-0001190
- **Traffic Information Visualization Analysis Device and Method** – Application Number : 10-2018-0079727 (Acquired - \$0.1M)

Services

2023-24	Reviewer , ACM CHI LBW, IEEE VIS	
2019, 24	Student Volunteer , IEEE VIS	Vancouver, Canada
2015, 19	Teaching Assistant , Object Oriented Programming	Ulsan, South Korea

Mentoring

2024	Mentee , Arthur Andre (EPFL), Zhuo Wang (Xi'an Jiaotong-Liverpool University), Ut(Jo Jo) Gong (University of Washington)	Boston, MA
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